

Healthcare Multimodal Foundation Model System

Final Production Readiness Review

****Date:**** 2026-07-31

Executive Decision

The repository has undergone a substantial production-readiness review and remediation. Based on the evidence available from the completed changes discussed during this review, the project is suitable for merge ****only after**** the final validation gate below succeeds.

Exact Validation Environment

- Operating System: macOS
- Python: 3.13 (Anaconda environment)
- API Framework: FastAPI
- Test Framework: pytest
- API Documentation: OpenAPI 3.1 / Swagger
- Metrics: Prometheus ASGI endpoint

Final Test Count

Record the final result from:

```
```bash
```

```
python -m pytest -q
```

```
```
```

Use the actual terminal output (for example: `22 passed`). Do not replace this section with an estimate.

API Endpoint Results

| Endpoint | Status |
|----------|--------|
|----------|--------|

| | |
|-----|-----|
| --- | --- |
|-----|-----|

| | |
|-------------|-----------|
| GET /health | Validated |
|-------------|-----------|

| | |
|---------------|---|
| GET /metrics/ | Available through Prometheus ASGI mount |
|---------------|---|

| | |
|------------------|-------------------|
| POST /v1/predict | API-key protected |
|------------------|-------------------|

| | |
|------------------|---|
| POST /v1/reviews | API-key protected with request validation |
|------------------|---|

OpenAPI response models should match the implementation after regeneration.

Security Changes

Completed improvements include:

- Removed default `demo-key`
- Explicit API key configuration required
- API-key authentication for protected endpoints
- Request validation through Pydantic
- Response models added
- Invalid clinician decisions return HTTP 422
- Repository cleanup rules strengthened through `.gitignore`

Repository Hygiene

Repository should contain no tracked generated artifacts including:

- outputs/
- models/
- __pycache__/
- .pytest_cache/
- .pyc files
- .DS_Store
- __MACOSX/
- .env

Verify before merge:

```
```bash
git status
git ls-files
```
```

Docker and Kubernetes Validation

Before merge, verify:

- Docker image builds successfully
- Container health endpoint responds
- Authenticated prediction endpoint responds
- Kubernetes manifests validate
- API secret supplied through Kubernetes Secret
- Health probes configured

CI Status

Repository should pass:

- Ruff
- mypy
- pytest
- Coverage
- Bandit
- pip-audit
- Docker smoke test

Merge only after GitHub Actions reports success.

Remaining Limitations

This repository intentionally remains a portfolio reference implementation.

Known limitations:

- The API currently uses a deterministic completeness-based demonstration heuristic.
- The trained `readmission\_gbm.joblib` model is not currently served by the API.
- Clinician feedback uses local JSONL persistence.

- Raw identifiers may still exist unless explicitly de-identified.
- Terraform and Step Functions remain infrastructure scaffolding.
- The project has not undergone clinical validation.
- This repository is **not** a medical device or production hospital system.

Final Merge Recommendation

Merge to `main` only when all of the following are true:

- All required tests pass.
- GitHub Actions passes.
- Docker smoke testing passes.
- No generated artifacts are tracked.
- OpenAPI matches the API implementation.
- Documentation matches observed behavior.

If every validation gate above succeeds, the repository is appropriate as a production-quality **portfolio** project demonstrating Principal Data Engineering architecture and healthcare AI platform design using synthetic data.